



## Chapter 2

# Oracle Architecture

CT208H – Oracle Database Management System



## Content

1. Oracle Database Architecture
2. Oracle Product Stack
3. Oracle Database Logical Architecture
4. Oracle Database Physical Architecture
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## 1. ORACLE DATABASE ARCHITECTURE

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➤ *Oracle Database Architecture*

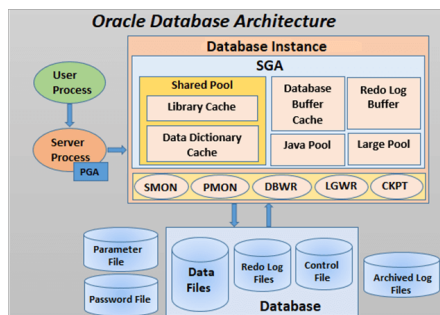
## Oracle Database Architecture

- **Oracle Database** employs a **client-server architecture**:
  - The **database server** and **client** run in separate processes
  - **Server process**: manages data and resources
  - **Client process**: communicates with the server to access and manipulate data
- SQL, PL/SQL, and Java are among the **programming interfaces** supported by Oracle Database
- It also **includes tools** for DB management and administration, such as Oracle Enterprise Manager and SQL Plus

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## Oracle Database Architecture

- The **Oracle Database architecture** consists of:
  - Database instances (memory structure)
  - Database system
  - Processes



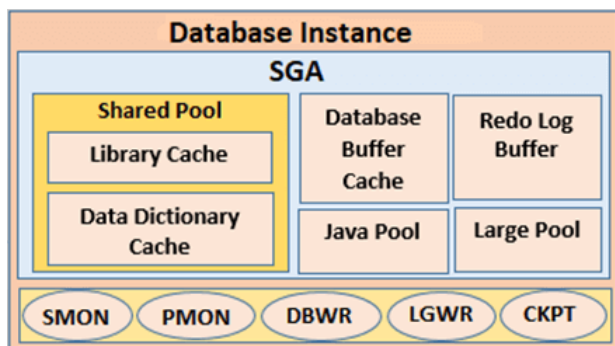
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## Oracle Instances

- The **instance** is a collection of **two things**:
  - System Global Area (SGA)
  - Background process
- It can be **started** and **stopped**
- When the running instance is **shut down**, all trace of its existence **goes away** at the same time
- The processes that make up the instance are known as **background processes**

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## Oracle Instances



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## System Global Area (SGA)

- A **shared memory area** of an Oracle instance
- When a **database instance starts**, SGA memory is **allocated**
- SGA stores **data** and **control information** about one database instance through its various **subcomponents**
- SGA contains three **mandatory** data structures:
  - The DB buffer cache
  - The log buffer
  - The shared pool

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## System Global Area (SGA)

- **Database Buffer cache:**
  - A portion of the SGA that stores **copies of data blocks** read from datafiles
  - It is used to **cache frequently accessed data blocks**, reducing the number of disks I/O required to access the data
  - This can **improve performance** by **reducing disk I/O time** and **increasing the speed** of data retrieval
  - The size of the database buffer cache **can be adjusted dynamically** and can be **automatically managed**

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## System Global Area (SGA)

- **Redo log buffer:**
  - The most crucial structure for **recovery operations**
  - The database's **guarantee** that data will **never be lost**
  - Consists of two or more pre-allocated files that **store all changes made** to the database as they occur
  - **Every instance** of an Oracle Database has an associated **redo log** to protect the database in case of an instance failure

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## System Global Area (SGA)

- **Shared pool:**
  - A portion of the SGA that contains **shared memory structures**, such as shared SQL and PL/SQL areas
  - It is used to store the **parsed representation** of SQL statements, **execution plans**, and **PL/SQL program units**
  - This allows for the efficient reuse of **frequently executed statements**, reducing the need for reparsing and **improving performance**

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## System Global Area (SGA)

- **Some other optional portions of SGA:**
  - **Large pool:**
    - It can be used for **large memory allocation**, such as backup and restore operations and I/O server processes
    - It is typically used to **improve the performance** of these operations by reducing the amount of **disk I/O** required
  - **Java pool:** used by Java Virtual Machine and related components to **run Java applications**

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## Background Process

- These processes are responsible for **managing memory**, performing **I/O operations**, and other **maintenance activities**
  - System Monitor Process (SMON): performs **system-level recovery** and **maintenance activities**
  - Process Monitor Process (PMON): monitors **other background processes**
  - Database Writer Process (DBWR): manages **data blocks**
  - Log Writer Process (LGWR): manages the **Redo blocks**
  - CheckPoint (CKPT): maintains **data files** and **control files**

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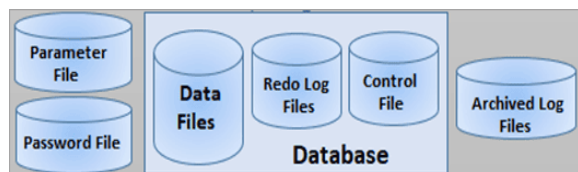
## Database System

- The Database system is suited to the storage system of a computer, and it is simply the **storage of files**
- Three categories** of files are situated in the database system:
  - Data files: hold the **actual data** in the database
  - Redo log files: hold **the changes** made in the database, used by the **recovery process** to retrieve the original information
  - Control files: hold **database status-related information** like database name, data file, and redo log file names, file locations, and log sequence number

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## Database System

- Other categories of files that contribute to DB management:
  - Parameter file**: contains the **parameters** that define the way the database is expected to **start up**
  - Password file**: holds the **user passwords** to maintain the **security** of databases



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## Processes

- There are **two types** of processes:
  - User process**:
    - Also known as the **client process** or **background process**
    - Helps the users **connect to the DB instance**
    - It is stabilized when the user sends a connection request to the Oracle server
  - Server process** (foreground process): connects the user to the database and **performs the activities on the client's behalf**, such as executing SQL statements or retrieving data from the DB

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## Processes

- User processes generate SQL; server processes execute SQL
- Each **server process** is associated with an area of **non-sharable memory**, called the **Program Global Area (PGA)**
  - This is a **private section** available only for server processes (while the SGA is available to all the foreground and background processes)
- **Background processes** also have PGA
- User processes usually connect to background processes through a LAN using the **Oracle Net protocol**

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## 2. ORACLE PRODUCT STACK

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## Oracle Server Family

- Three **primary groupings** of products within the server technology family:
  - **Database server**
  - **Application server**
  - **Enterprise manager suites**
- These form the basic components for Oracle's vision of **grid computing**

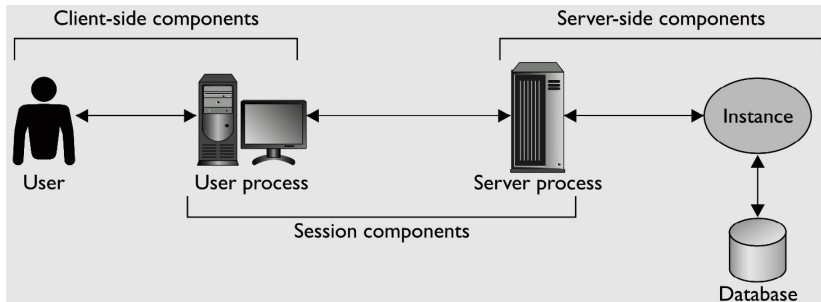
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## Database Server

- Comprises **Oracle instances** and **databases** with many features like Streams, Partitioning, Warehousing, Replication, and RAC
- Processing model: **client/server (2-tier)**:
  - The **client tier** consists of two components: the **users** and the **user processes**
  - The **server tier** has three components: the **server processes** that execute the SQL, the **instance**, and the **database** itself

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## Database Server



*The indirect connection between a user and a database*

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## Database Server

- An Oracle database is a **set of files** on disk
- **Access** to the database is through the **Oracle instance**
- The instance is a set of **processes** and **memory structures**
  - It exists on the **CPU(s)** and in the **memory** of the **server node**
  - Its existence is **temporary**
- **Users** of the database **establish sessions** against the instance, which then **manages all access** to the database

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## Application Server

- Oracle **Application Server** provides a **platform** for developing, deploying, and managing **web applications**
- An application server allows client-side software, traditionally installed on **end-user computers**, to be replaced by applications hosted and executing from a **centralized location**
- The application **user interface** is commonly exposed to users via their **web browsers**
- These applications may use **one or more database servers**

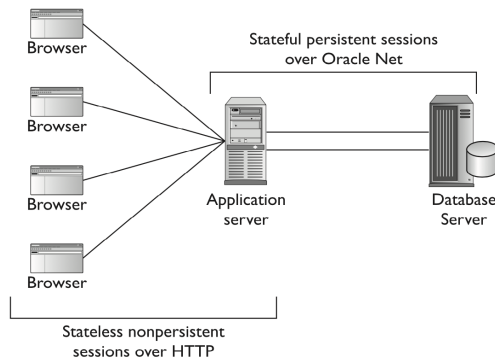
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## Application Server

- **Web applications** usually run in **at least three tiers**:
  - A **database tier** manages access to the data,
  - The **client tier** (often implemented via a web browser) handles the local window management for communications with the users
  - An **application tier** in the middle executes the program logic that generates the user interface and the SQL calls to the database

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## Application Server



*The connection pooling model*

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## Enterprise Manager

- Oracle **Enterprise Manager** comes in **three forms**:
  - **Database Control**: a graphical tool for managing **databases**
  - **Application Server Control**: a graphical tool for managing one or more **application server instances**
  - **Grid Control**: control globalizes the management **environment**. It enables administrators to manage hundreds/thousands of targets
- Database Control and Application Server Control consist of a **Java process** running on the server machine, which listens for **HTTP** or **HTTPS** connection requests

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## Oracle Development Tools

- Oracle provides several **tools** for:
  - **Developing applications**: Oracle Developer Suite (Forms, Reports, and Discoverer), Oracle Application Server,
  - **Utility programs**: Oracle E-Business Suite, Oracle Collaboration Suite,...
  - **Languages**: SQL, PL/SQL, Java. Oracle also provides the OCI (Oracle Call Interface) libraries that let code written in these languages connect to an Oracle DB and invoke SQL commands

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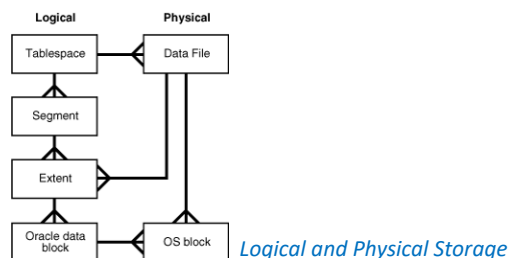
## 3. ORACLE DATABASE LOGICAL ARCHITECTURE

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## Logical storage structure

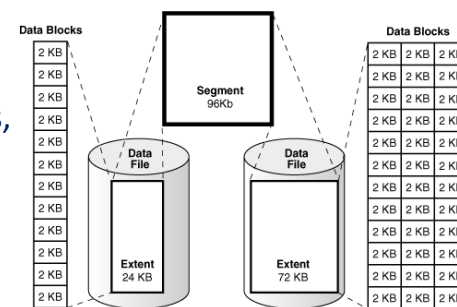
- Oracle Database allocates **logical space** for all data in the database
- Logical units:
  - Data blocks
  - Extents
  - Segments
  - Tablespaces
- The data in the **data files** is stored in **operating system blocks**



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## Logical Storage Hierarchy

- A **segment** contains one or more **extents**, each of which contains multiple **data blocks**
- The figure shows the relationships among data blocks, extents, and segments within a tablespace
  - The **segment** has **two extents** stored in different data files



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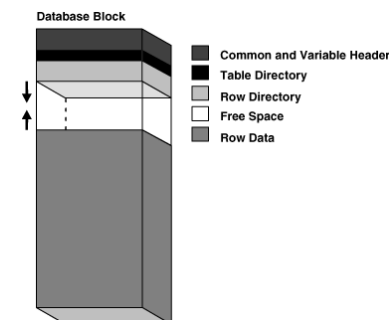
## Data blocks

- A data block (or Oracle block/Page) is the **minimum unit** of database I/O
- At the **physical level**, database data is stored in **disk files** made up of **operating system blocks**
- Oracle block is a **logical storage structure** whose size and structure are **not known** to the operating system
- The **standard data block size** for a database is 4KB or 8KB (it can be set and must be a multiple of the operating system block size)

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## Data blocks

- Every data block has a format or **internal structure** that enables the database to track the data and free space in the block
- Oracle can use **table compression** to eliminate duplicate values in a data block



Data Block Format

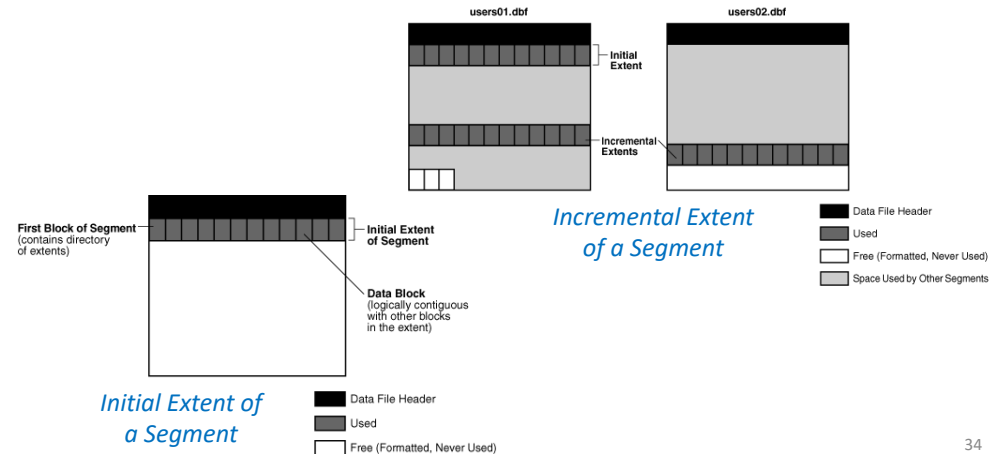
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## Extents

- An extent is a unit of database storage made up of logically contiguous data blocks
- Data blocks can be physically spread out on disks because of RAID striping and file system implementations
- By default, the database allocates an initial extent for a data segment when the segment is created
- An extent is always contained in one data file

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## Extents



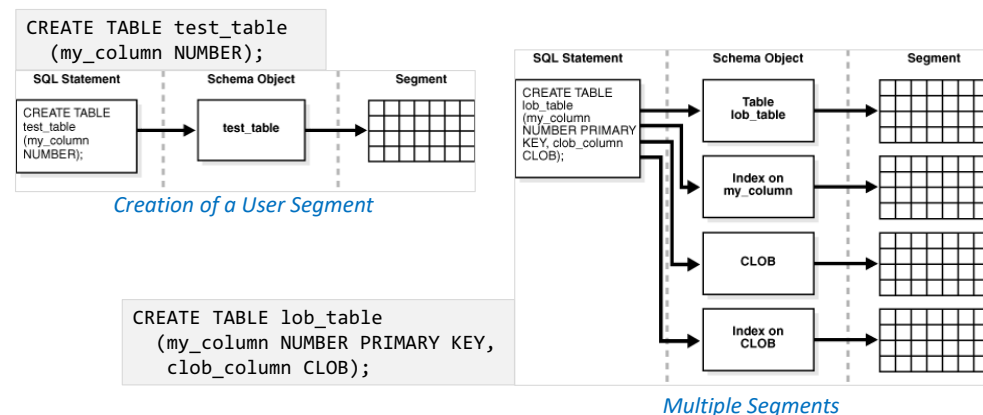
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## Segments

- A segment is a set of extents that contains all the data for a logical storage structure within a tablespace
- For example, Oracle Database allocates one or more extents to form the data segment for a table
- The database also allocates one or more extents to form the index segment for an index on a table

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## Segments



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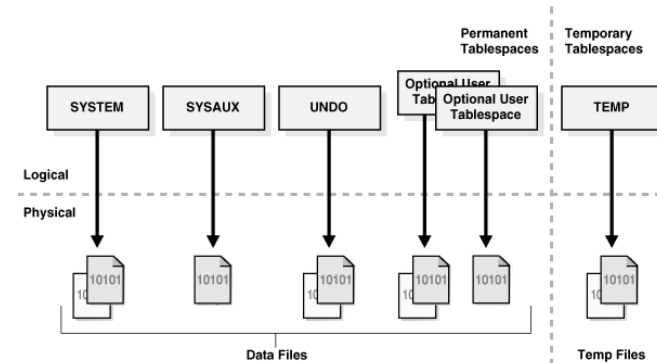
## Tablespaces

- A tablespace is a **logical storage container** for segments
  - Segments are **database objects**, such as tables and indexes, that consume storage space
  - At the **physical level**, a tablespace **stores data** in one or more **data files** or temp files
- A database must have the **SYSTEM** and **SYSAUX** tablespaces
- Each **database user** is assigned a default permanent tablespace

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## Tablespaces

- The tablespaces in a typical database



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## Tablespaces

- **Permanent** tablespaces:
  - Groups **persistent** schema objects
  - The segments for objects in the tablespace are stored physically in data files
  - Each **database user** is assigned a **default** permanent tablespace
  - It is recommended that we should create **at least one tablespace** to store user and application data

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## Tablespaces

- The **SYSTEM** tablespace:
  - An **administrative tablespace**, included with the database
  - The **SYSTEM** tablespace includes the following **information**, all owned by the **SYS** user:
    - The **data dictionary**
    - **Tables and views** that contain administrative information about the database
    - **Compiled stored objects** such as triggers, procedures
    - It cannot be **renamed** or **dropped**

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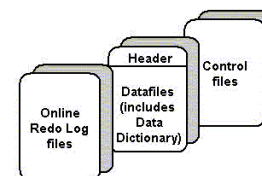
## Tablespaces

- The SYSAUX tablespace:
  - An **auxiliary** tablespace to the **SYSTEM** tablespace
  - Helps to **reduce the load** and **enhance the performance** of the SYSTEM tablespace
  - Contains **many important Oracle system** components such as Oracle Enterprise Manager, Oracle Streams,...
  - Helps to **increase security** and **control access** to important system data by **separating** management components into SYSAUX

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## Database Files

- An **Oracle instance** consists of several **physical files**:
  - **Datafiles**: contain the **actual data** of the databases
  - **Redo log files**: contain a **chronological record of changes** made to the database and **enable recovery** when failures occur
  - **Control files**: used to synchronize all **database activities**



## 4. ORACLE DATABASE PHYSICAL ARCHITECTURE

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## Other Key Files

- **Parameter file:** there are two types
  - The `init.ora` file (or PFILE): a **static parameter file** that contains parameters that specify how to **start up a database instance**
  - The `spfile.ora`: a **dynamic parameter file** that stores parameters to specify how to **start a database** (its parameters can be modified while the database is running)
- **Password file:** which users are **authenticated to startup/shutdown** an Oracle instance
- **Archived redo files:** copies of the redo log files

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## 5. CONNECT TO ORACLE DATABASE

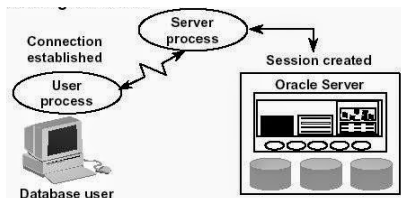
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➤ *Connect to Oracle database*

## Connect to Oracle Database

- To connect to Oracle Database, we need a **client program**, such as **SQL\*Plus** or **SQL Developer**
- When connected, we can **view schema objects**, the **properties** and **data** of Oracle Database tables, and use **queries** to retrieve data from Oracle Database tables

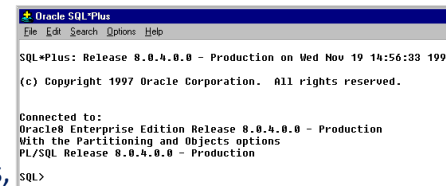


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## Connect to Oracle from SQL\*Plus

- To connect to Oracle Database from SQL\*Plus:
  1. Open the Windows **command prompt**.
  2. Type `sqlplus` and then press Enter
  3. Type your **user name**
  4. Type your **password**

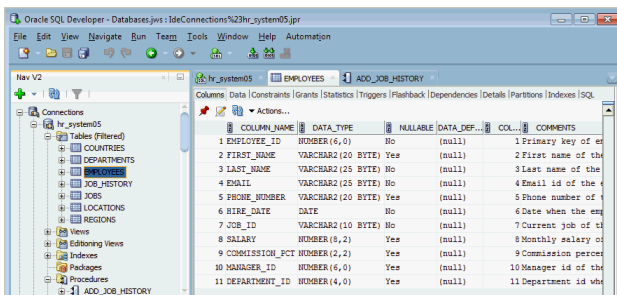
At the **SQL>** prompt, you can enter and run SQL\*Plus commands, SQL statements, PL/SQL statements, and operating system commands



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## ORACLE DB Management Connect to Oracle from SQL Developer

- Oracle SQL Developer is a **graphical version** of SQL\*Plus that gives database developers a **convenient** way to perform basic tasks



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## ORACLE DB Management Connect to Oracle from SQL Developer

- This tool is **included** in the Oracle Database installation
- Installer: <http://www.oracle.com/technetwork/developer-tools/sql-developer/>
  - Unzip the SQL Developer file into a folder (directory) of your choice, which will be referred to as **<sqldeveloper\_install>**
  - Run **sqldeveloper.exe** in the **<sqldeveloper\_install>\sql developer** folder
- Oracle SQL Developer installation **guide**:  
<https://docs.oracle.com/en/database/oracle/sql-developer/18.2/rptig/installing-sql-developer.html#RPTIG-GUID-A3509B92-90A4-4268-8027-FE85DE5554A8>

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## ORACLE DB Management

## 6. ORACLE DATABASE SQL EXECUTION

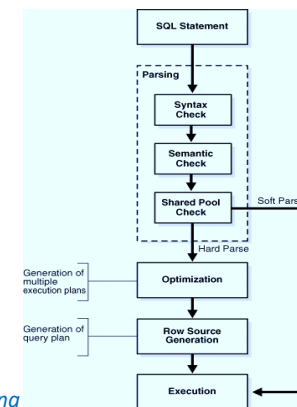
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## ORACLE DB Management

## About SQL processing

- SQL processing** is the **parsing**, **optimization**, **row source generation**, and **execution** of a SQL statement
- The figure depicts the **general stages** of SQL processing
- Depending on the statement, the database **may omit some** of these stages



Stages of SQL Processing

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## SQL Parsing

- The **first state** of SQL processing
- Involves **separating** the pieces of a SQL statement **into a data structure** that other routines can process
- The applications make a **parse call** to the database to prepare the statement for execution
- The parse call creates a **cursor** which is a handle for the **session-specific private SQL area** that holds a parsed SQL statement and other processing information

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## SQL Parsing

- Syntax check:**
  - Check the **syntactic validity**
  - A statement that **breaks a rule for well-formed SQL syntax** **fails** the check
  - For example, the following statement **fails** because the keyword FROM is **misspelled** as FORM:

```
SQL> SELECT * FORM employees;
SELECT * FORM employees
      *
ERROR at line 1:
ORA-00923: FROM keyword not found where expected
```

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## SQL Parsing

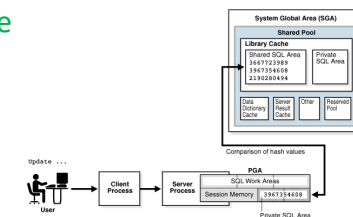
- Semantic check:**
  - The **semantics** of a statement are its **meaning**
  - A semantic check determines whether a statement is **meaningful**
  - For example, whether the objects and columns in the statement exist

```
SQL> SELECT * FROM nonexistent_table;
SELECT * FROM nonexistent_table
      *
ERROR at line 1:
ORA-00942: table or view does not exist
```

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## SQL Parsing

- Shared Pool check:**
  - To determine whether it **can skip resource-intensive steps** of statement processing
  - When a user submits an SQL statement, the database searches the shared SQL area to see if **an existing parsed statement** has the same **hash value**



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## SQL Optimization

- Improves the **efficiency** and **performance** of SQL queries to ensure SQL statements execute as quickly and efficiently as possible
- Just for **DML** statements
- Exception** only occurs when the **DDL** includes a **DML** component, such as a **subquery** that requires optimization

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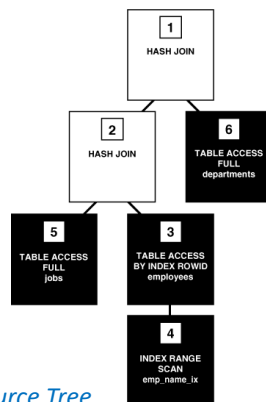
## SQL Row Source Generation

- A software that receives the **optimal execution plan** from the optimizer and produces an **iterative execution plan** that is usable by the rest of the database
- Iterative execution plan:
  - The plan takes the form of a **combination of steps**
  - Each step returns a **row set**
  - A row source is a **row set** returned by a step in the execution plan, along with a **control structure** that can **iteratively process** the rows

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## SQL Execution

- SQL engine executes each **row source in the tree** produced by the **row source generator**
  - The order of the steps in execution is the **reverse** of the order in the plan
  - Each node of the tree acts as a **row source** (each step of the execution plan either retrieves rows from the database or accepts rows from one or more row sources as input)



Row Source Tree

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## QUESTIONS?

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